B.M.S. College of Engineering

Bio-inspired Algorithms | 23DS7BSBIO

Concrete Compressive Strength Prediction Using Differential Evolution–Optimized Support Vector Regression

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Introduction to Optimization Problems

Objective and search space

- **Objective:** minimize the five-fold development RMSE of an RBF-SVR.
- **Decision variables:** $\log_{10} C$, $\log_{10} \gamma$ and $\log_{10} \varepsilon$.
- **Constraints:** fixed parameter bounds and a budget of 300 objective calls.
- **Evaluation:** every candidate means fitting models; no derivatives are available.
- **Holdout:** used only to score fixed selections after tuning.

At a glance

3
decision variables

300
objective calls per run

5
CV folds per evaluation

What are Nature-Inspired Algorithms?

Nature-inspired search

- **Idea:** search methods motivated by biological, collective or physical processes.
- **Mechanism:** explore candidate solutions; measured fitness guides the updates.
- **Genetic Algorithm:** selection, crossover and mutation.
- **Differential Evolution:** mutation from population-vector differences.
- **Particle Swarm:** individual and social search information.
- **Caveat:** no gradient is required, but performance depends on task and controls.

At a glance

0
gradients required

1
method implemented: DE

3
models compared

Classification of Nature-Inspired Algorithms

Algorithm families

- **Evolutionary:** Genetic Algorithms and Differential Evolution.
- **Swarm:** Particle Swarm Optimization and Ant Colony Optimization.
- **Physical process:** Simulated Annealing.
- **Implemented:** DE/rand/1/bin for continuous parameters.
- **Comparator:** Random Search with an equal tuning budget.
- **Background only:** the other algorithms are not measured baselines.

At a glance

3
families covered

1
implemented: DE/rand/1/bin

1
comparator: Random Search

Problem Statement

Prediction task

- **Task:** predict concrete compressive strength (MPa) from ingredients and age.
- **Difficulty:** SVR accuracy depends on C, kernel γ and tolerance ε.
- **Question:** does DE improve tuning under an equal evaluation budget?
- **Comparison:** default SVR, Random Search–SVR and DE–SVR.
- **Controls:** splits, preprocessing, bounds and budgets stay fixed.
- **Reporting:** all predefined seeds and the limits of the comparison.

At a glance

8
input features

1
target: strength (MPa)

3
models compared

Objectives and Scope

Research questions

- **RQ1:** does tuning improve a default RBF-SVR? (frozen study)
- **RQ2:** does DE beat equally budgeted Random Search? (frozen study)
- **RQ3:** how much does the input representation matter? (Extension 1, ln(age))
- **RQ4:** does DE keep an edge in a 10-D search space and on new splits? (Extension 2)
- **Scope:** one UCI dataset, one RBF-SVR, explicit DE/rand/1/bin; extensions were pre-registered.

At a glance

4
research questions

2
pre-registered extensions

5
optimizer seeds

Dataset and Preprocessing

Dataset and evaluation split

- **Source:** UCI Concrete Compressive Strength; no missing values.
- **Inputs:** seven ingredient quantities (kg/m³) and curing age (days).
- **Target:** compressive strength (MPa).
- **Grouping:** 996 unique input tuples; identical inputs share a group.
- **Validation:** five group-disjoint CV folds; StandardScaler fitted within each fold.
- **Features:** frozen runs use the original inputs; Extension 1 adds ln(age).

At a glance

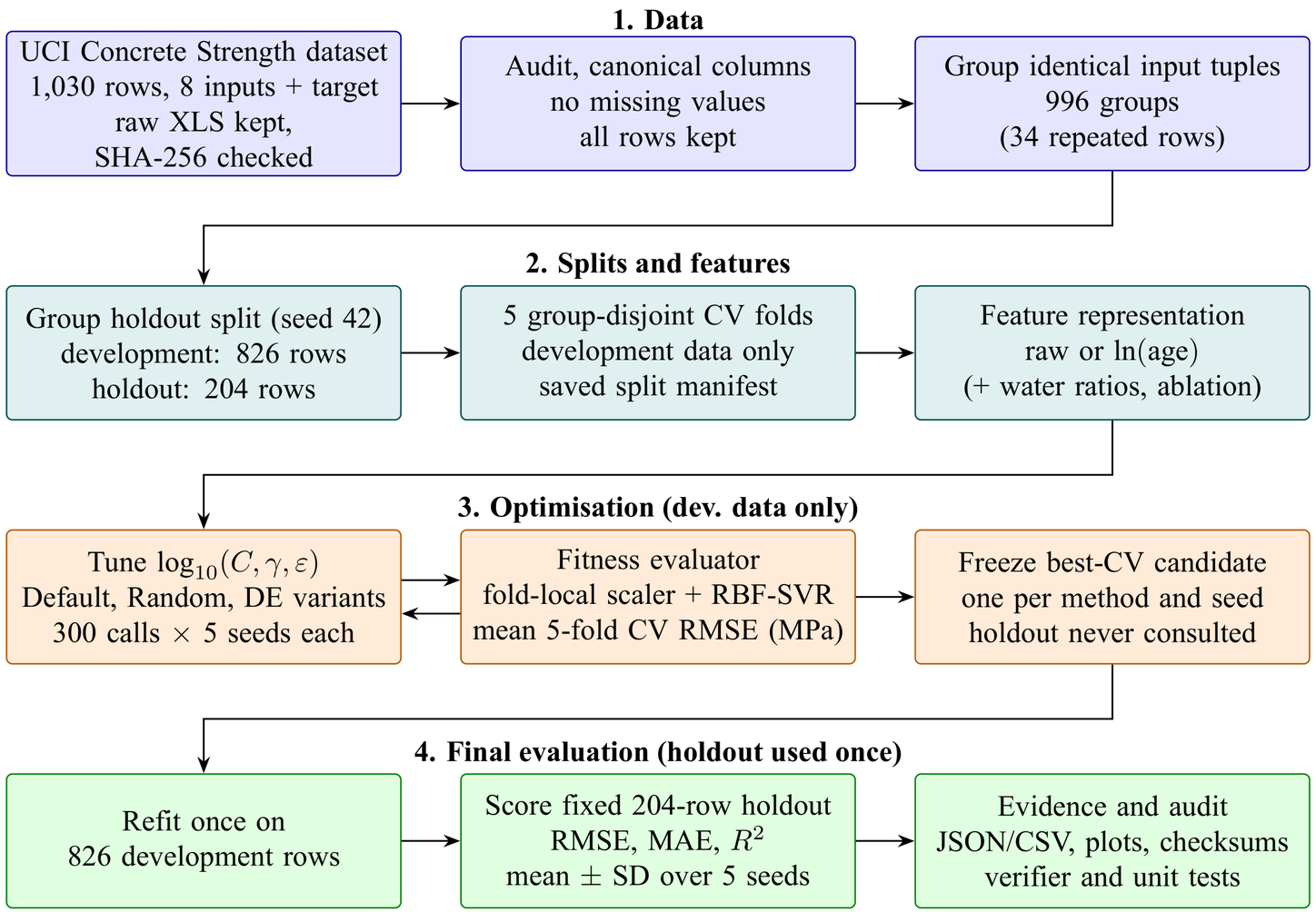
1,030
rows

8
input features

826 / 204
development / holdout rows

796 / 200
development / holdout groups

System Architecture: Data to Evidence



Four stages: data, splits and features, development-only optimisation, single holdout evaluation.

Implementation and Results

Pipeline

- **Data:** raw-data audit and saved group splits.
- **Model:** StandardScaler → RBF-SVR.
- **Search:** development CV evaluates every candidate.
- **Refit:** selected parameters refit on all development rows.
- **Evaluate:** fixed holdout scored once.
- **Save:** histories, predictions and provenance.

DE configuration

15 population	**0.8 / 0.9** F / CR
300 calls incl. initialization	**19** trial generations
5 optimizer seeds	**[−2, 4]** $\log_{10}$ C bounds
[−5, 1] $\log_{10}$ γ bounds	**[−3, 1]** $\log_{10}$ ε bounds

Differential Evolution: Explicit Operators

DE/rand/1/bin

1. **Initialize:** 15 candidates uniformly within the log bounds.
2. **Donors:** three distinct vectors, all excluding the current target.
3. **Mutate:** $v = x_{r1} + 0.8(x_{r2} - x_{r3})$, clipped to bounds.
4. **Crossover:** binomial at 0.9; one mutant coordinate is forced.
5. **Evaluate:** mean development CV RMSE of the trial.
6. **Select:** keep the trial if no worse; updates deferred to generation end.
7. **Stop:** after exactly 300 objective calls; no hidden polishing.

Settings

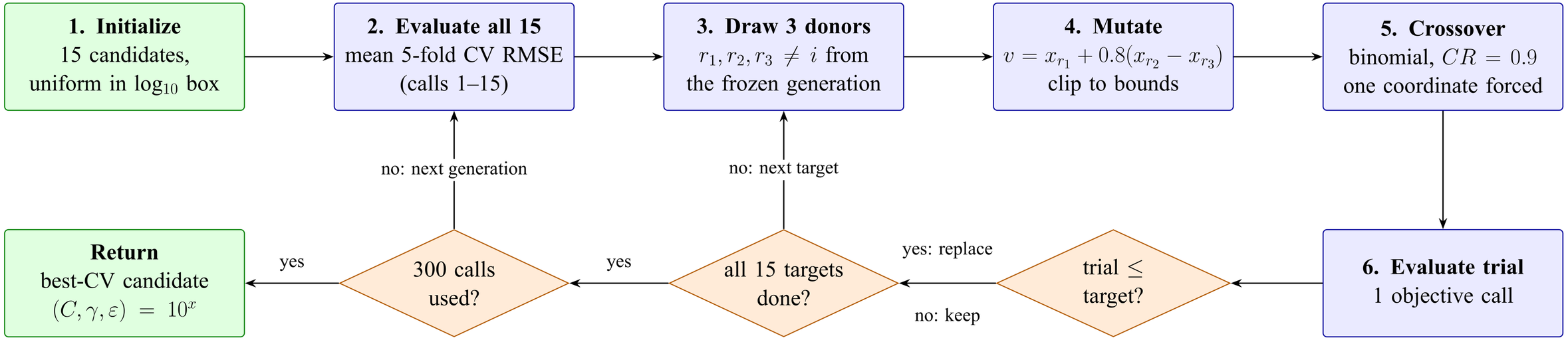
F = 0.8
mutation weight

CR = 0.9
crossover rate

15
population size

300
objective calls per seed

DE Flowchart (as Implemented)



Per generation
15 trial candidates; donors from the frozen generation

Acceptance
greedy: trial replaces target if no worse

Budget
15 + 19 generations × 15 trials = 300 calls per seed

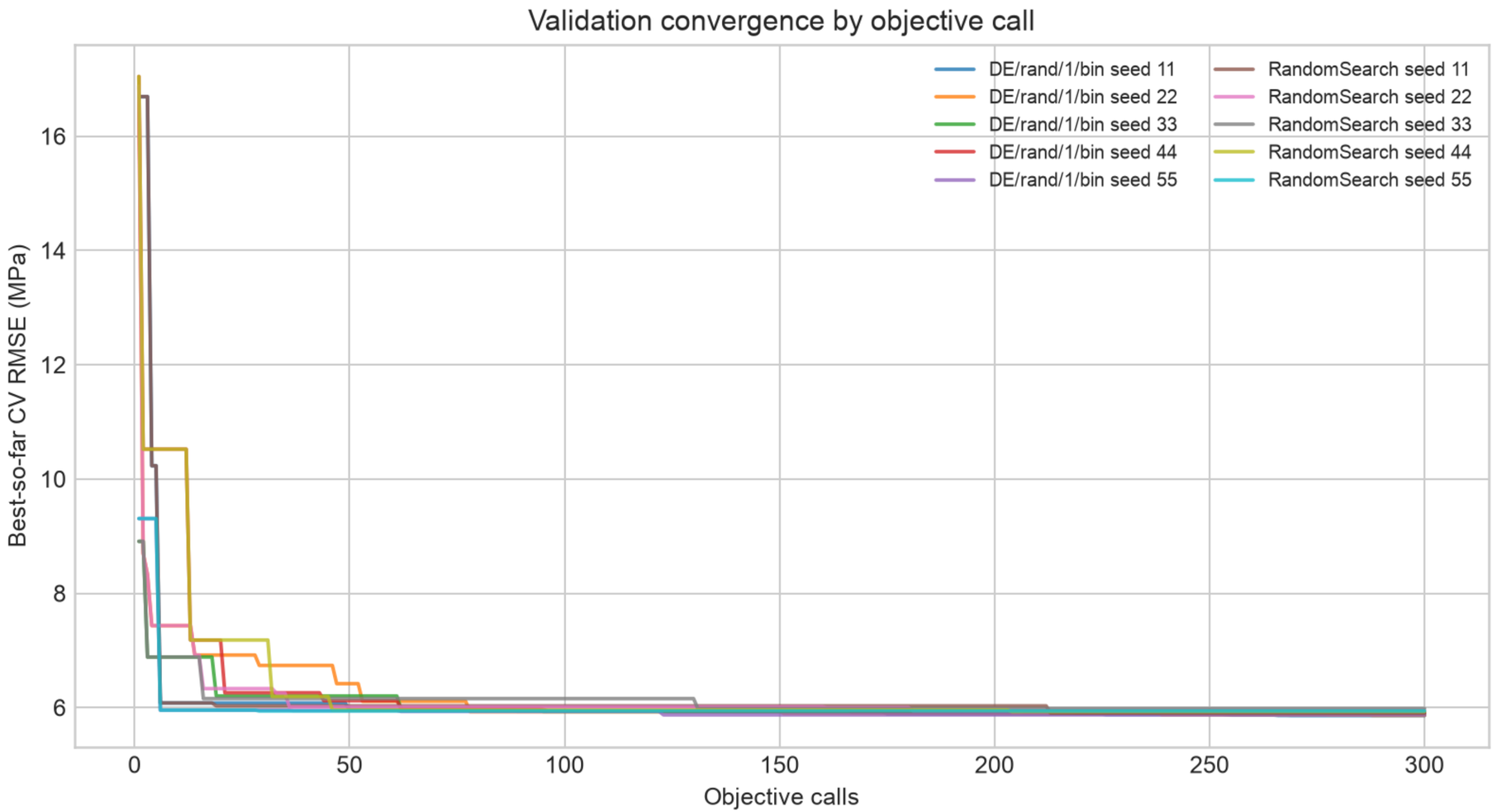
Performance Metrics and Benchmarking

Same 204 holdout rows; tuned values are mean ± sample SD over five seeds.

Model	RMSE (MPa)	MAE (MPa)	R^2
Default SVR	9.9069	7.8854	0.5755
Random Search–SVR	6.2328 ± 0.1194	4.3502 ± 0.0253	0.8319 ± 0.0064
DE–SVR	**6.2250 ± 0.1104**	**4.3143 ± 0.0526**	**0.8324 ± 0.0060**

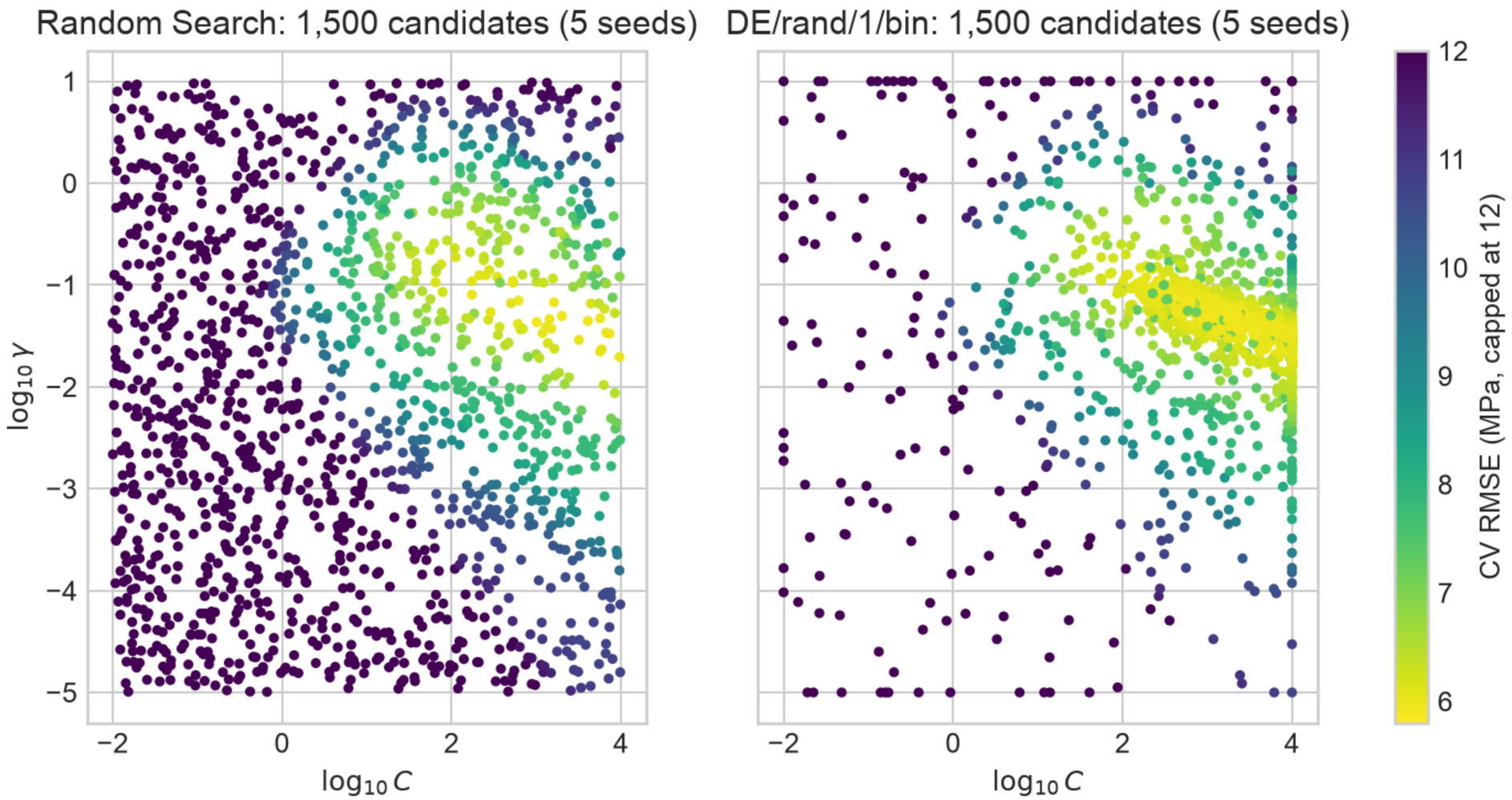
- **Improvement:** DE reduces RMSE by about 37.2% relative to default SVR.
- **Gap:** DE and Random Search differ by only 0.0078 MPa (about 0.1%).
- **Conclusion:** tuning helps; the two searches are essentially equivalent here.

Development Validation Convergence



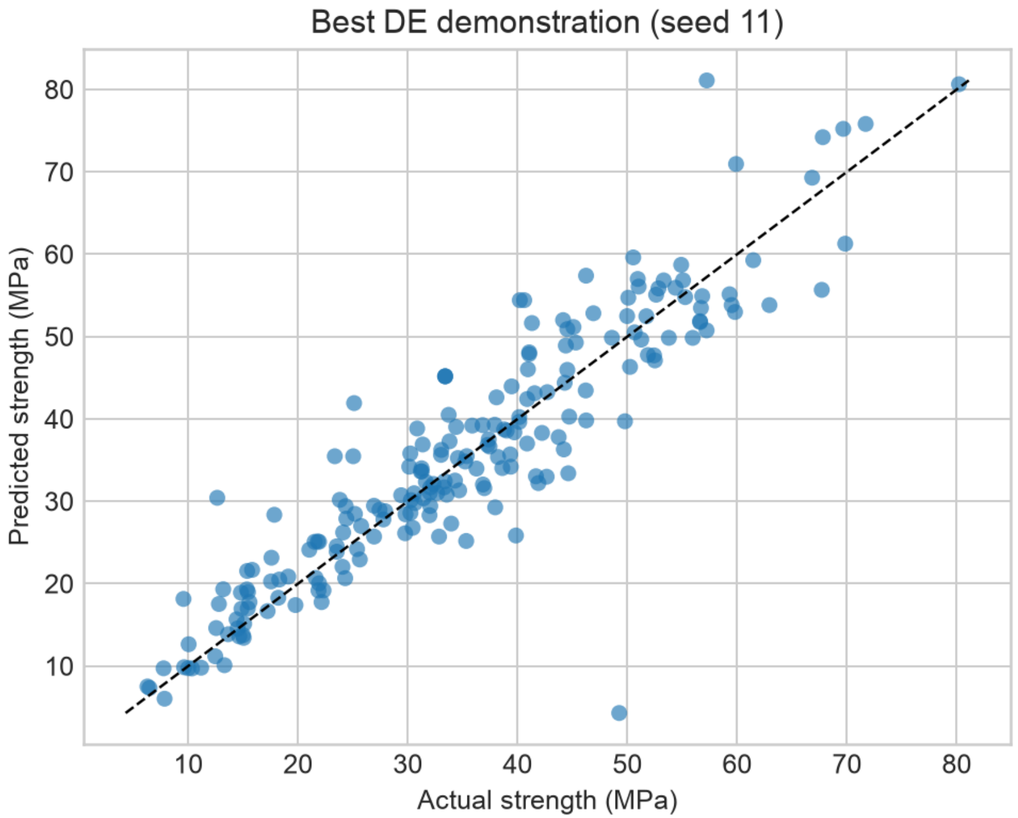
Equal budgets: 300 calls per method per seed, including initialization.

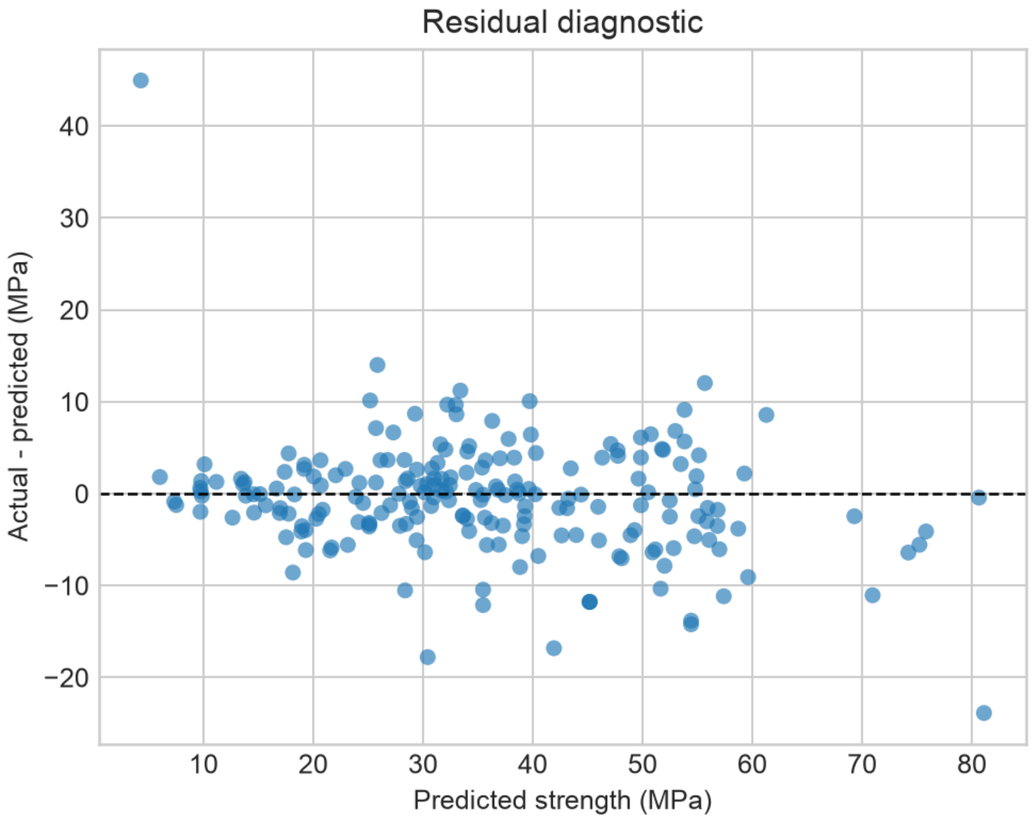
Where Each Search Evaluated Candidates



1,500 candidates per method (5 seeds): DE concentrates in the good region; Random Search stays uniform.

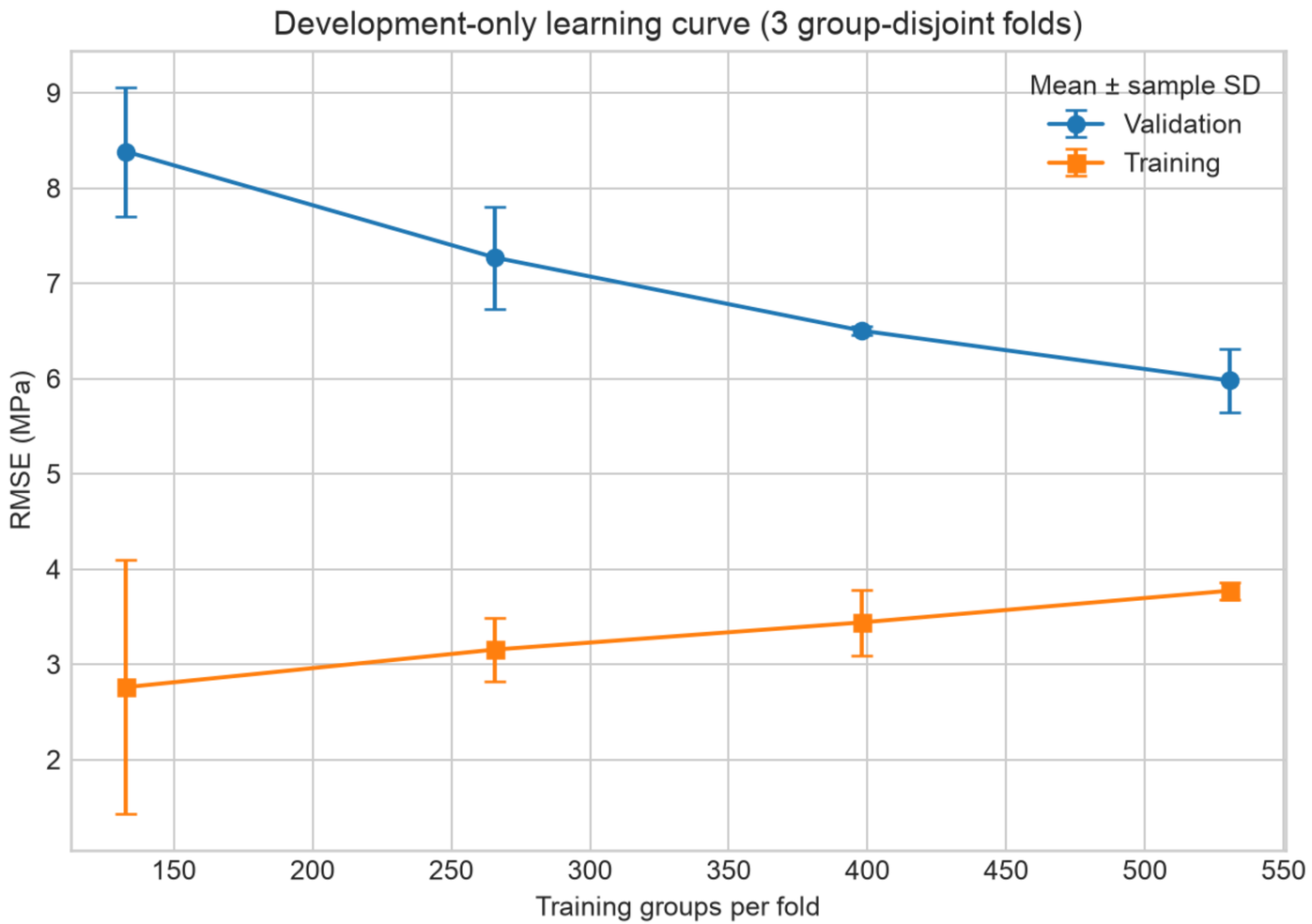
Predictions and Residuals





Demonstration: DE seed 11, chosen by validation; holdout RMSE 6.3712 MPa.

Learning-Curve Diagnostic



Three development-only folds; nested whole groups; fixed parameters; no retuning.

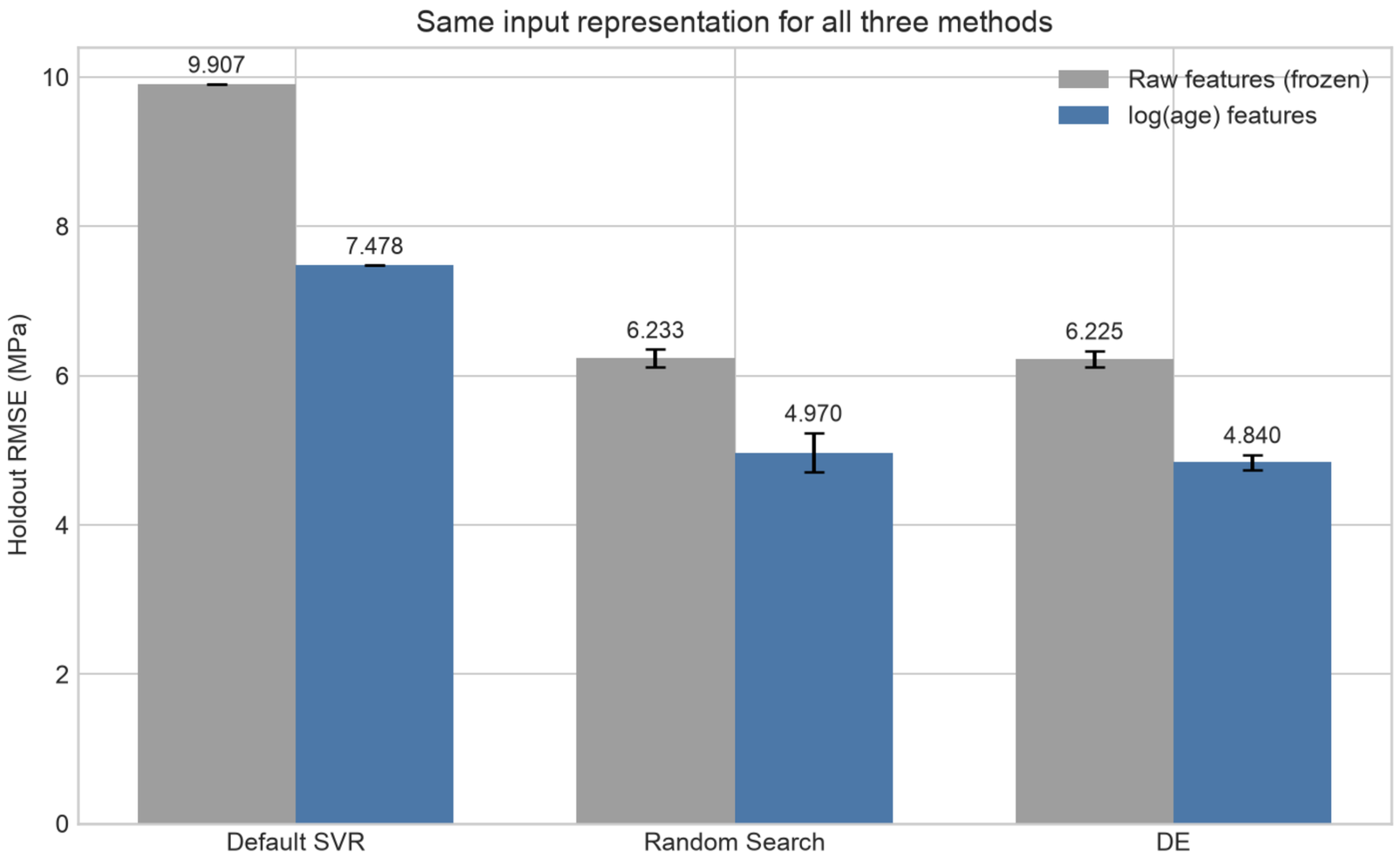
Extension: ln(age) Input for All Three Methods

Same 204 holdout rows; tuned values are mean ± sample SD over five seeds.

Model	Raw RMSE (MPa)	ln(age) RMSE (MPa)	ln(age) R^2
Default SVR	9.9069	7.4780	0.7581
Random Search–SVR	6.2328	4.9700 ± 0.2656	0.8929
DE–SVR	6.2250	4.8402 ± 0.1039	0.8987

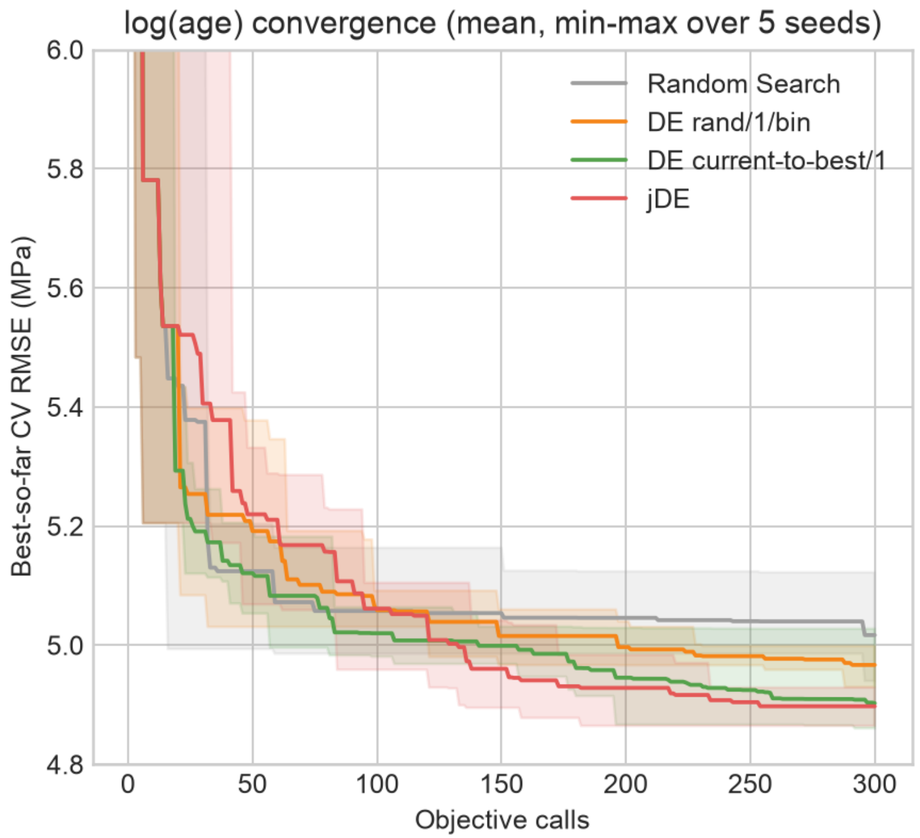
- **Representation:** DE 6.225 → 4.840 MPa (22.2% lower).
- **DE vs Random Search:** 0.130 MPa apart on ln(age); no significance claim.
- **Disclosure:** ln(age) was chosen after a development-only probe; the holdout was inspected in Stage 1.

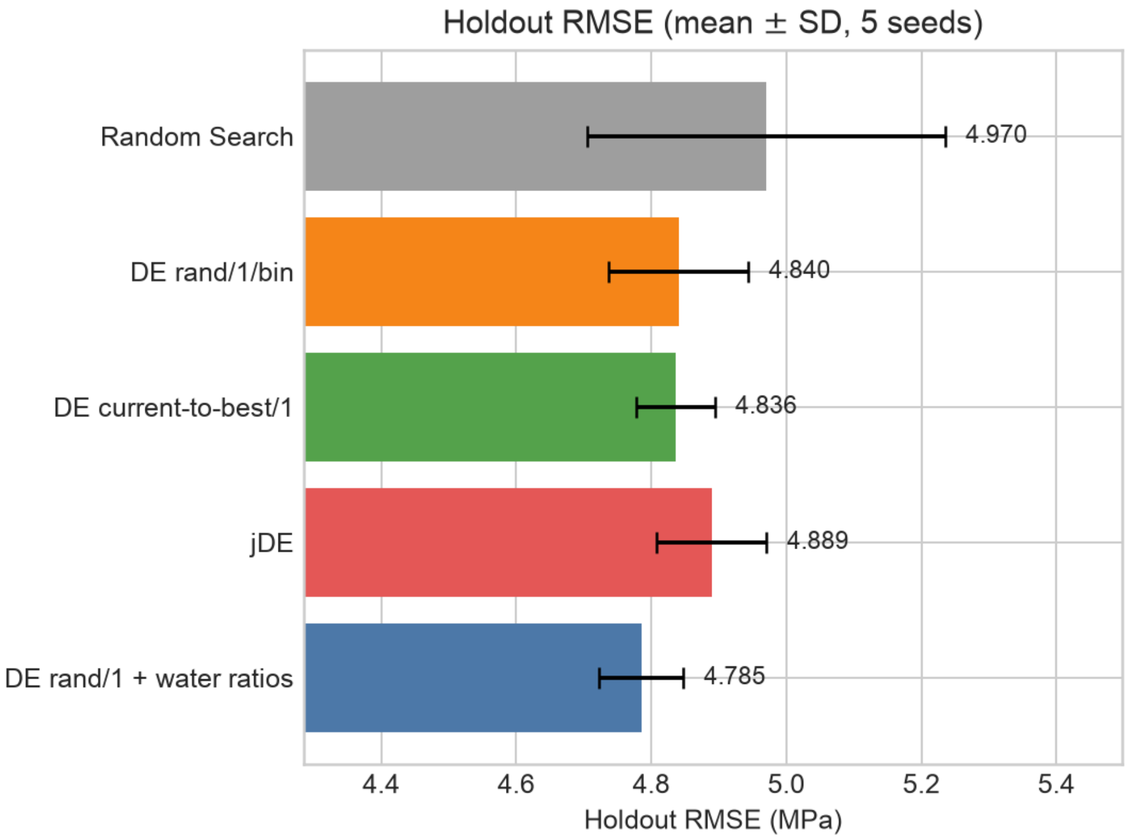
Representation Comparison



Holdout RMSE, mean ± SD over five optimizer seeds; raw rows are the frozen results.

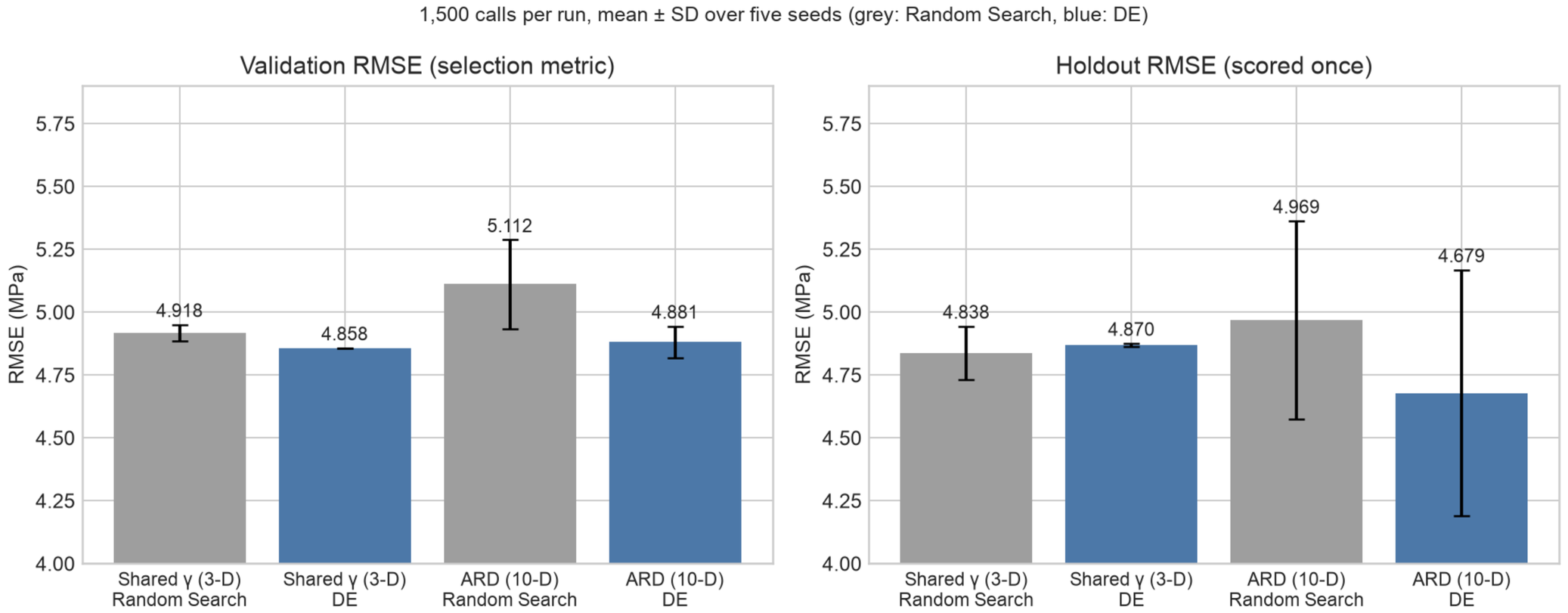
DE Variant Ablation (ln(age))





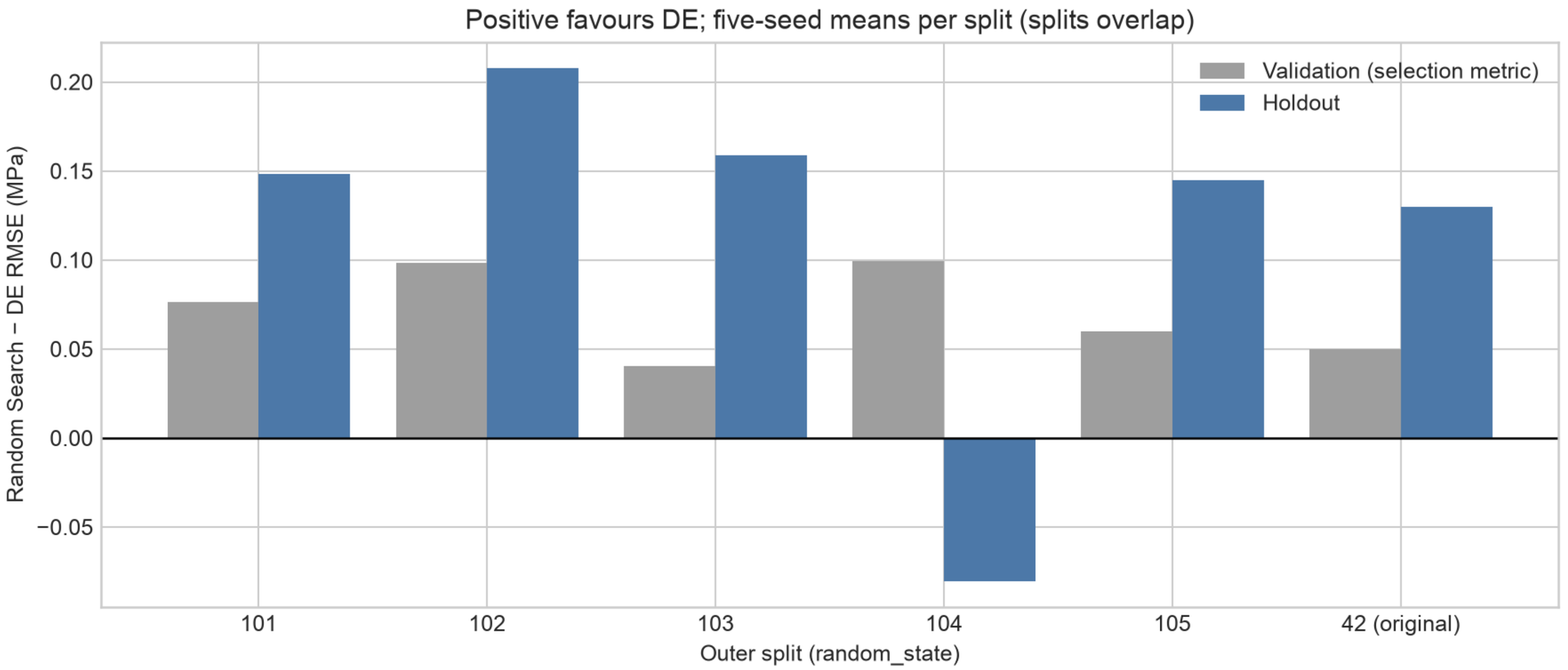
Same 300-call budget; rand/1/bin vs current-to-best/1/bin vs jDE vs Random Search.

Extension 2a: ARD Kernel (10-D Search Space)



1,500 calls, five seeds; y-axis starts at 4.0 MPa. ARD does not improve validation RMSE over a shared γ.

Extension 2b: Replication on Five New Splits



DE lower validation RMSE on 6/6 splits, holdout on 5/6; mean holdout gap 0.118 MPa (indicative; splits overlap).

Key Findings

Summary of evidence

- **Tuning:** DE cuts holdout RMSE 37.2% (9.907 → 6.225 MPa).
- **Frozen features:** DE and Random Search are 0.0078 MPa apart.
- **Representation:** ln(age) lowers DE to 4.840 MPa (22.2%), the largest effect.
- **DE edge:** about 0.1 MPa, on 5 of 6 splits; no significance claim.
- **ARD (10-D):** no accuracy gain; Random Search degrades there, DE does not.

Key numbers

37.2%
lower RMSE than default SVR

22.2%
lower DE RMSE with ln(age)

≈0.1 MPa
DE edge over Random Search

5 of 6
splits favour DE on holdout

Advantages

What this approach offers

- **Transparent search:** explicit, tested DE operators for a three-parameter continuous search.
- **Clear gain:** both tuned methods substantially improve default SVR.
- **Leakage control:** group-aware splits and fold-local scaling.
- **Auditable budget:** equal candidate budgets for every search.
- **Reproducible:** complete histories, predictions and locked dependencies.
- **Verified:** 57 passing tests, checksummed evidence and a saved audit.

Evidence

57
passing tests

3
checksummed evidence sets

300
calls per search, equal

5
seeds per method

Limitations

What the evidence cannot establish

- **Data:** one public dataset; one main holdout (Extension 2 adds five overlapping splits).
- **Uncertainty:** seed SD shows optimizer variability, not dataset uncertainty.
- **DE vs Random Search:** no meaningful superiority is demonstrated.
- **Chronology:** baseline and Random Search holdout evaluations preceded final DE evaluation.
- **Analysis:** no feature-importance or significance study; ln(age) and ARD came from development probes.
- **Scope:** a finite search cannot prove global optimality; no field deployment is validated.

Scale of evidence

1
dataset

204
main holdout rows

5
optimizer seeds

None
formal significance test

Future Scope

Next experiments

- **External data:** evaluate a frozen model on external laboratory data.
- **Splits:** non-overlapping outer splits or nested group cross-validation.
- **DE controls:** test settings systematically under matched budgets.
- **Uncertainty:** study age-stratified errors with a declared protocol.
- **Interpretation:** add studies without using holdout feedback for tuning.
- **Impact:** assess material or environmental benefits only with suitable new evidence.

Priorities

External
laboratory data

Nested
group cross-validation

Systematic
DE control study

Declared
protocol before evaluation

References

Selected sources (click to open)

[1] I.-C. Yeh, Concrete Compressive Strength, UCI Machine Learning Repository, 1998.

[2] I.-C. Yeh, Cement and Concrete Research, 28(12), 1797–1808, 1998.

[3] R. Storn and K. Price, Journal of Global Optimization, 11, 341–359, 1997.

[4] J. Bergstra and Y. Bengio, JMLR, 13, 281–305, 2012.

[5] A. J. Smola and B. Schölkopf, Statistics and Computing, 14, 199–222, 2004.

[6] scikit-learn documentation: SVR, StandardScaler and GroupKFold.

[7] Project repository: code, results and report.

Thank You

Tuning helps
DE cuts holdout RMSE 37.2% versus default SVR

Representation matters most
ln(age) lowers DE RMSE by 22.2%

DE's edge is small
about 0.1 MPa; untested for significance

Questions and discussion

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Guide: Prof. Manoj Kumar, Assistant Professor

github.com/R-V-Abhishek/concrete-strength-de-svr